Unit: Unit 0: Foundations of Algebra **Name:** ____________
Topic: Isolating Variables **Date:** ________

Quiz (Carolina Reaper)

Q1. Solve for x in the equation $2x + 5 = 11$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q2. A truck is traveling from City A to City B at an average speed of x miles per hour. If the distance between the two cities is 240 miles and the travel time is 4 hours, what is the value of x?
(A) $x = 50$
(B) $x = 60$
(C) $x = 70$
(D) $x = 80$
(E) $x = 90$

Q3. Solve for y in the equation $y - 3 = 7$
(A) $y = 4$
(B) $y = 5$
(C) $y = 10$
(D) $y = 11$
(E) $y = 12$

Q4. A logistics company has a fleet of x trucks, and each truck can carry 15 crates. If the company needs to transport 315 crates, what is the value of x?
(A) $x = 15$
(B) $x = 20$
(C) $x = 21$
(D) $x = 25$
(E) $x = 30$

Q5. Solve for z in the equation $z + 2 = 9$
(A) $z = 5$
(B) $z = 6$
(C) $z = 7$
(D) $z = 10$
(E) $z = 11$

Q6. A ship is traveling from Port A to Port B at an average speed of x knots. If the distance between the two ports is 180 nautical miles and the travel time is 6 hours, what is the value of x?
(A) $x = 20$
(B) $x = 25$
(C) $x = 30$
(D) $x = 35$
(E) $x = 40$

Q7. Solve for w in the equation $w - 2 = 11$
(A) $w = 9$
(B) $w = 10$
(C) $w = 12$
(D) $w = 13$
(E) $w = 14$

Q8. A traffic flow model is given by the equation $x + 2y = 12$, where x is the number of cars and y is the number of trucks. Solve for y when $x = 4$
(A) $y = 2$
(B) $y = 3$
(C) $y = 4$
(D) $y = 5$
(E) $y = 6$

Open Ended Questions (FRQ)

Q9. Solve for x in the equation $3x + 2 = 2x + 5$ and show all steps

Q10. A delivery company has a formula to calculate the delivery time t in hours, given the distance d in miles and the speed s in miles per hour: $t = \frac{d}{s} + 1$. Solve for s when $t = 3$ and $d = 120$

Chef's Tips

- Tip 1: Emphasize the importance of isolating variables in algebraic equations
- Tip 2: Use real-world examples from transportation and logistics to illustrate the concepts
- Tip 3: Encourage students to show all steps when solving equations